14 JULY 2021

Meet Johan

11 · Bad Neuenahr-Ahrweiler

Fictional character



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European Commission JOINT RESEARCH CENTRE

01 Am I in danger?

02 How serious is it?

03 Where should I go?

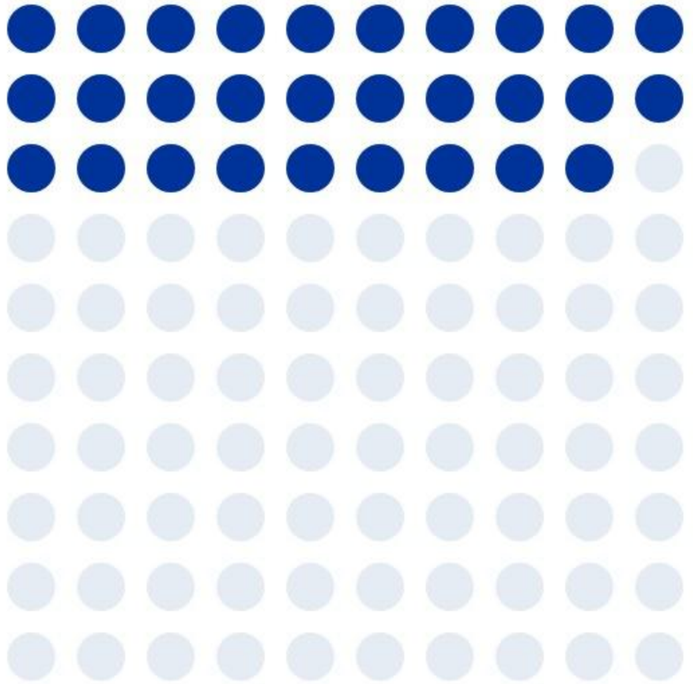


The water came at night

29%

received no warning

Respondents in Rhineland-Palatinate



Each dot = 1 percentage point

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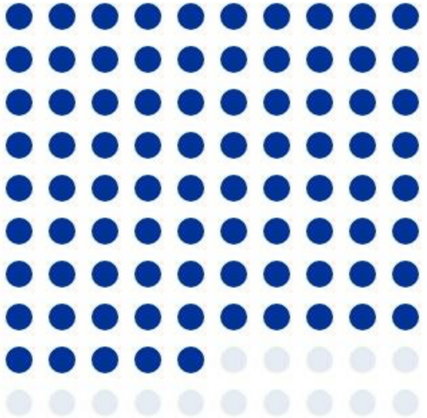
Thieken et al. (2023), NHESS 23, 973-990, DOI 10.5194/nhess-23-973-2023. July 2021 floods; online survey, n = 1,315 flood-affected respondents. 29%: Rhineland-Palatinate. 85% and 46%: warned respondents in Rhineland-Palatinate and North Rhine-Westphalia.

Among those warned

Warned respondents across both states

85%

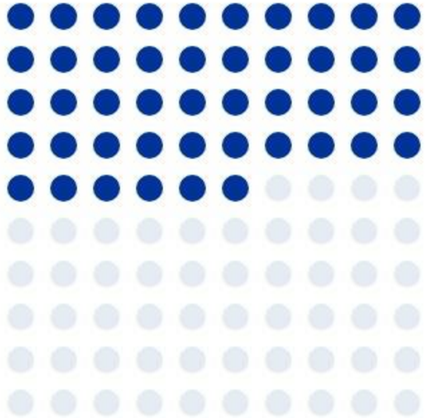
didn't expect very severe flooding



Share among warned respondents

46%

lacked knowledge on protective action



Share among warned respondents

Each grid represents a separate percentage. Overlap between the groups is not shown.

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From flood map to one clear message.

01 02 03



Flood + roads

→ closed roads

Copernicus EMS · EFAS · OpenStreetMap

Supports, and does not replace, official warnings and evacuation decisions.

From flood map to one clear message.

01

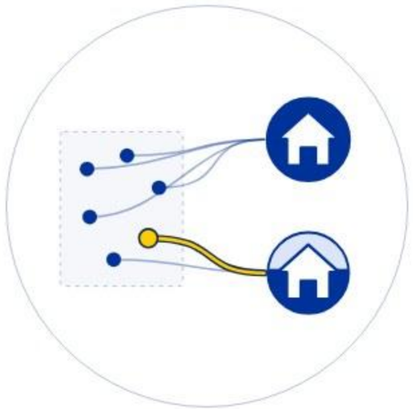


Flood + roads

→ closed roads

Copernicus EMS · EFAS · OpenStreetMap

02



Match + route

→ a shelter with room

GHS-POP · building capacity · open roads

03

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From flood map to one clear message.

01

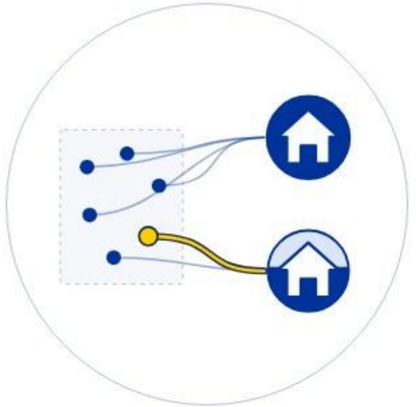


Flood + roads

→ closed roads

Copernicus EMS · EFAS · OpenStreetMap

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Match + route

→ a shelter with room

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Johan's alert

→ three answers, plain text

Cell Broadcast · works on 2G

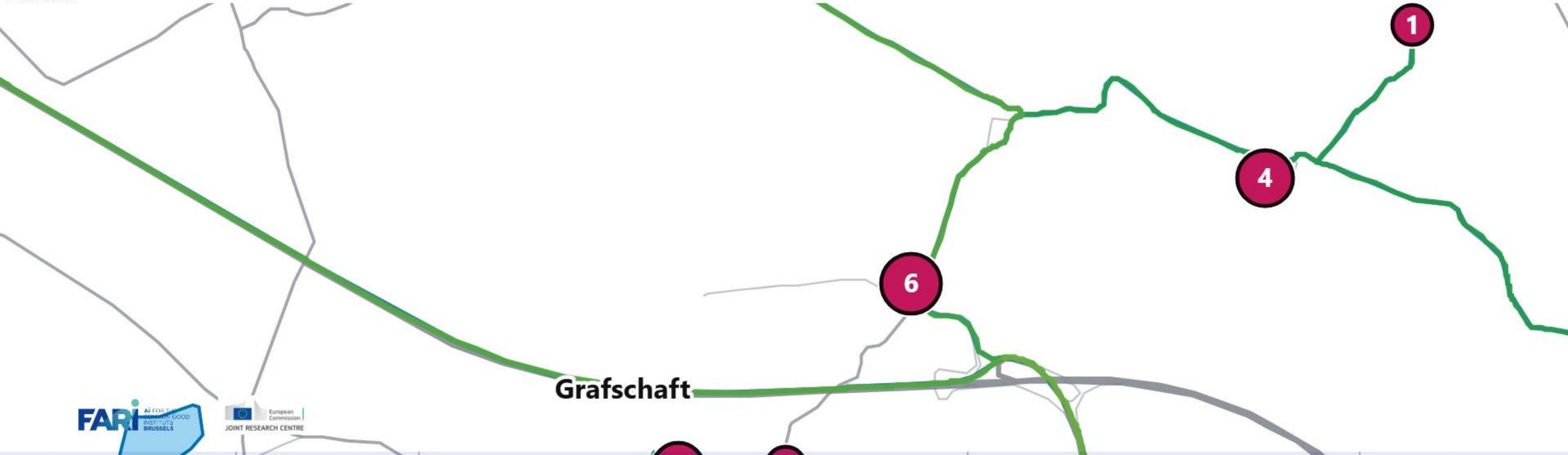
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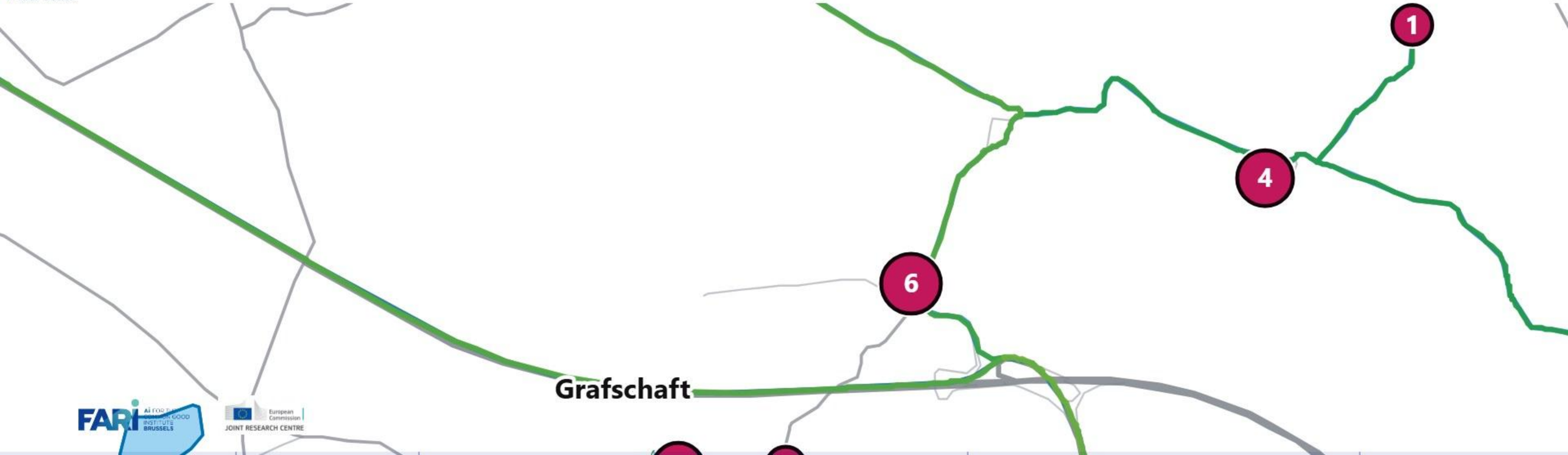
1. 01
Am I in danger?
2. 02
How serious is it?
3. 03
Where should I go?

2G
Saturday 31 October
06:12
!
Emergency Alertnow
Flood danger. Rüstringer Straße is in the flood area.
Severity: as in July 2021.
Leave now. Go to: Kindergarten, Vehner Weg 7.Walk 1 min to the nearest open road, then 1 min by car via Johannisstraße, Ehlinger Straße.
After your street, this route avoids flooded roads.
Demo message, not a real alert
Your route



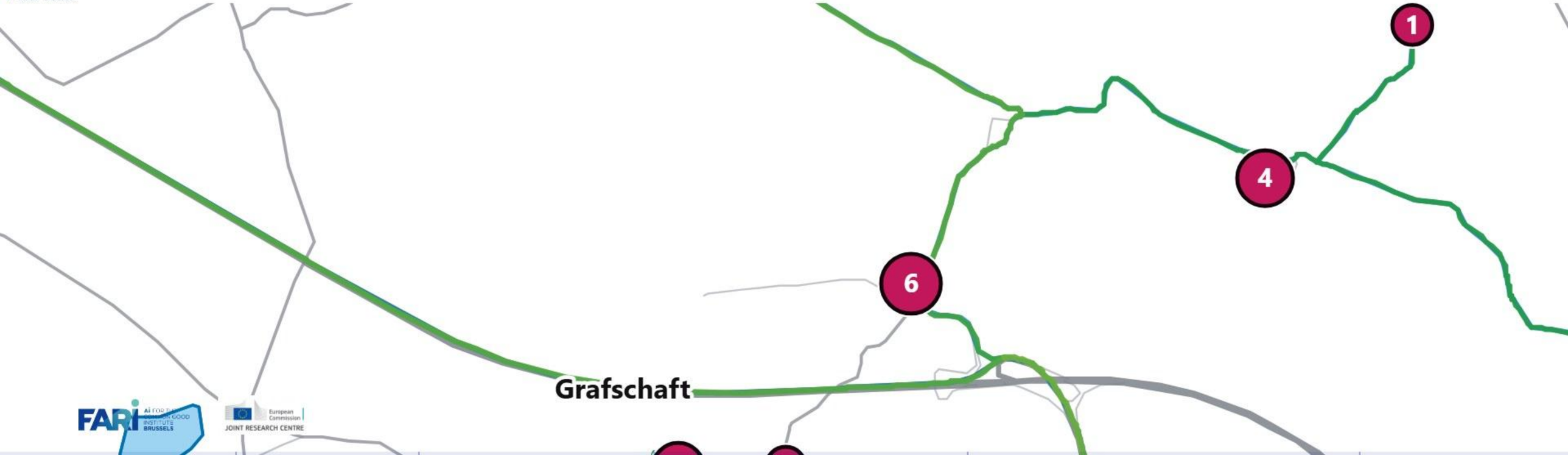
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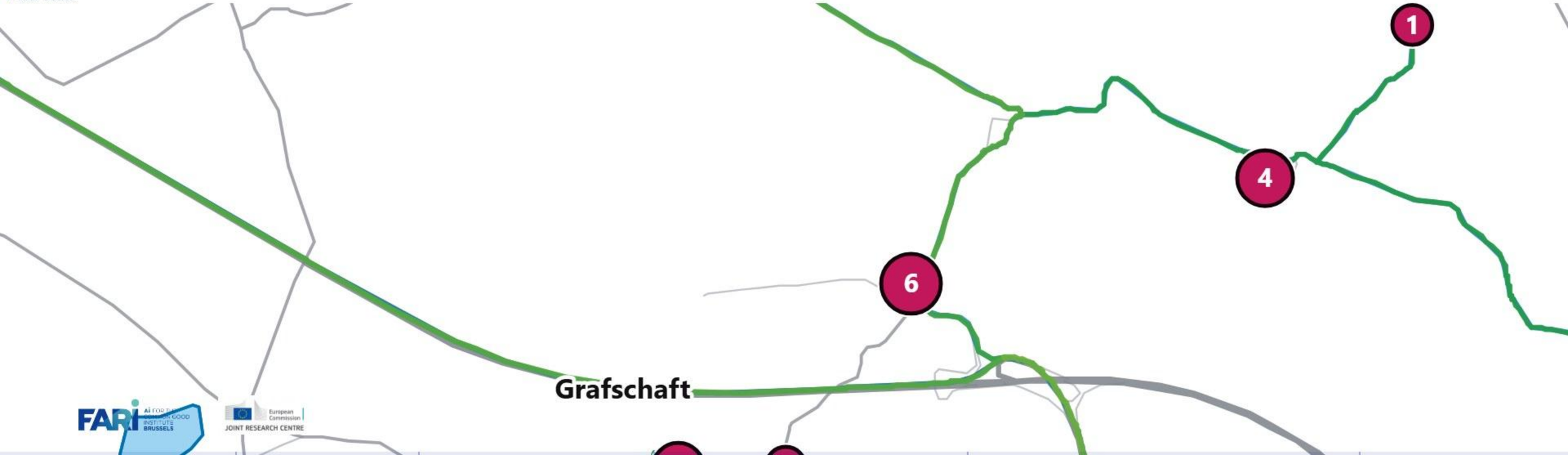
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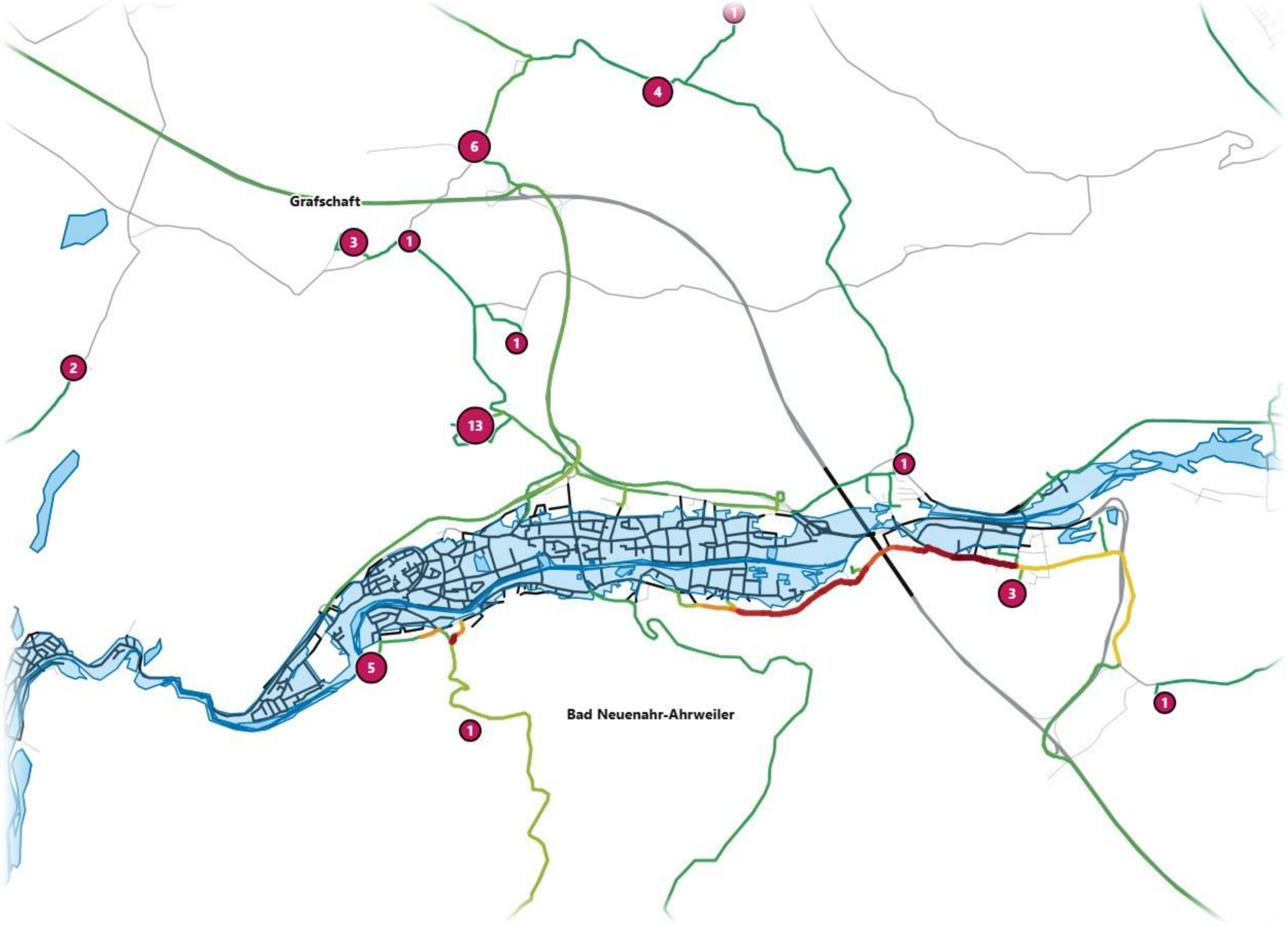


Streets close. Exits overload.

Flood, July 2021

Closed road

Road stress, model



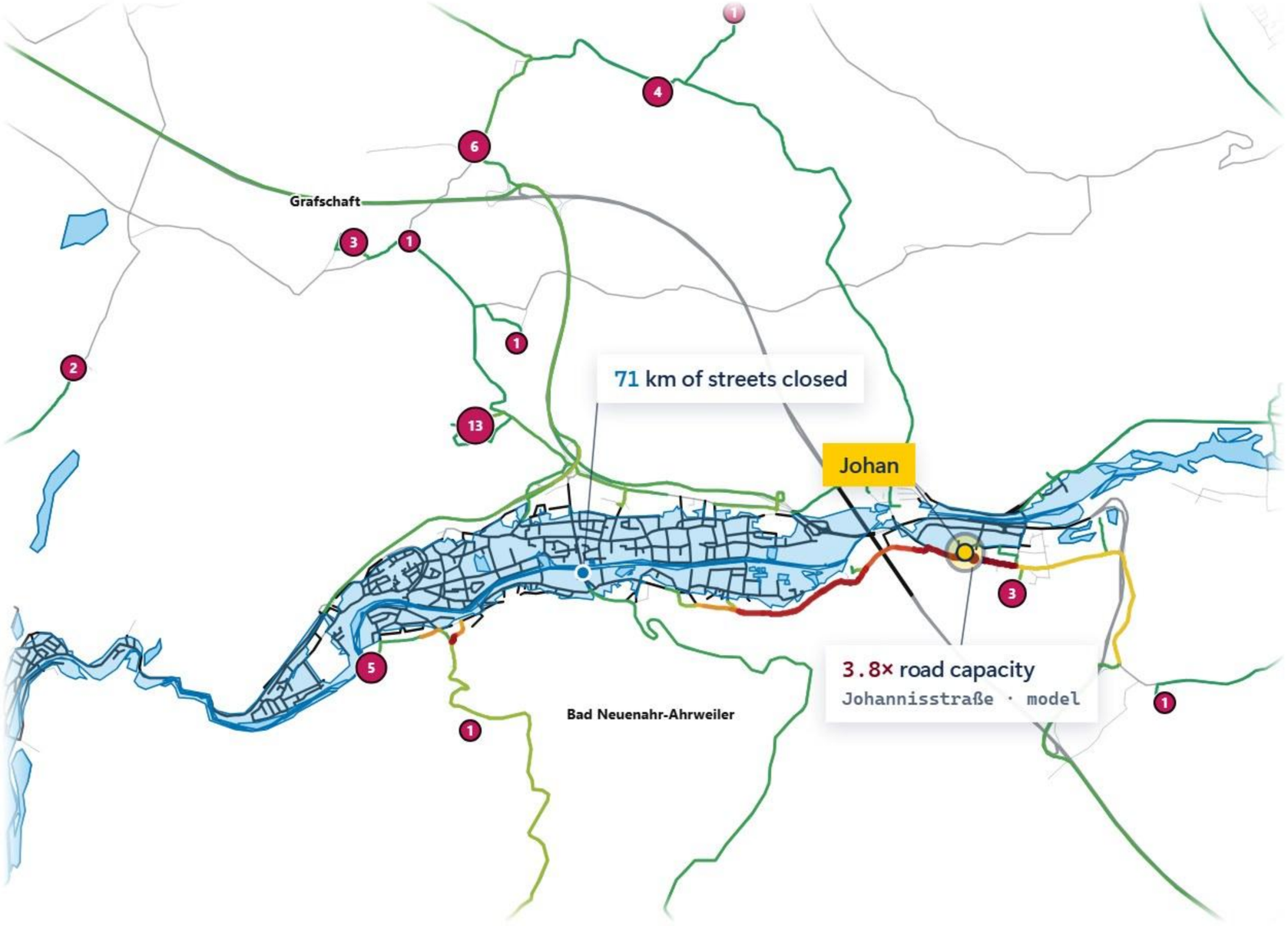
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Smart Shelter System · bundle 2026-10-30 09:46 UTC (commit 01abbb0) · flood: Copernicus EMS EMSR517 · residents: GHS-POP 2020 estimate · roads © OpenStreetMap contributors · capacity, car share and speeds are assumptions; stress is an indicator, not a forecast

Streets close. Exits overload.

- Flood, July 2021
- Closed road
- Road stress, model



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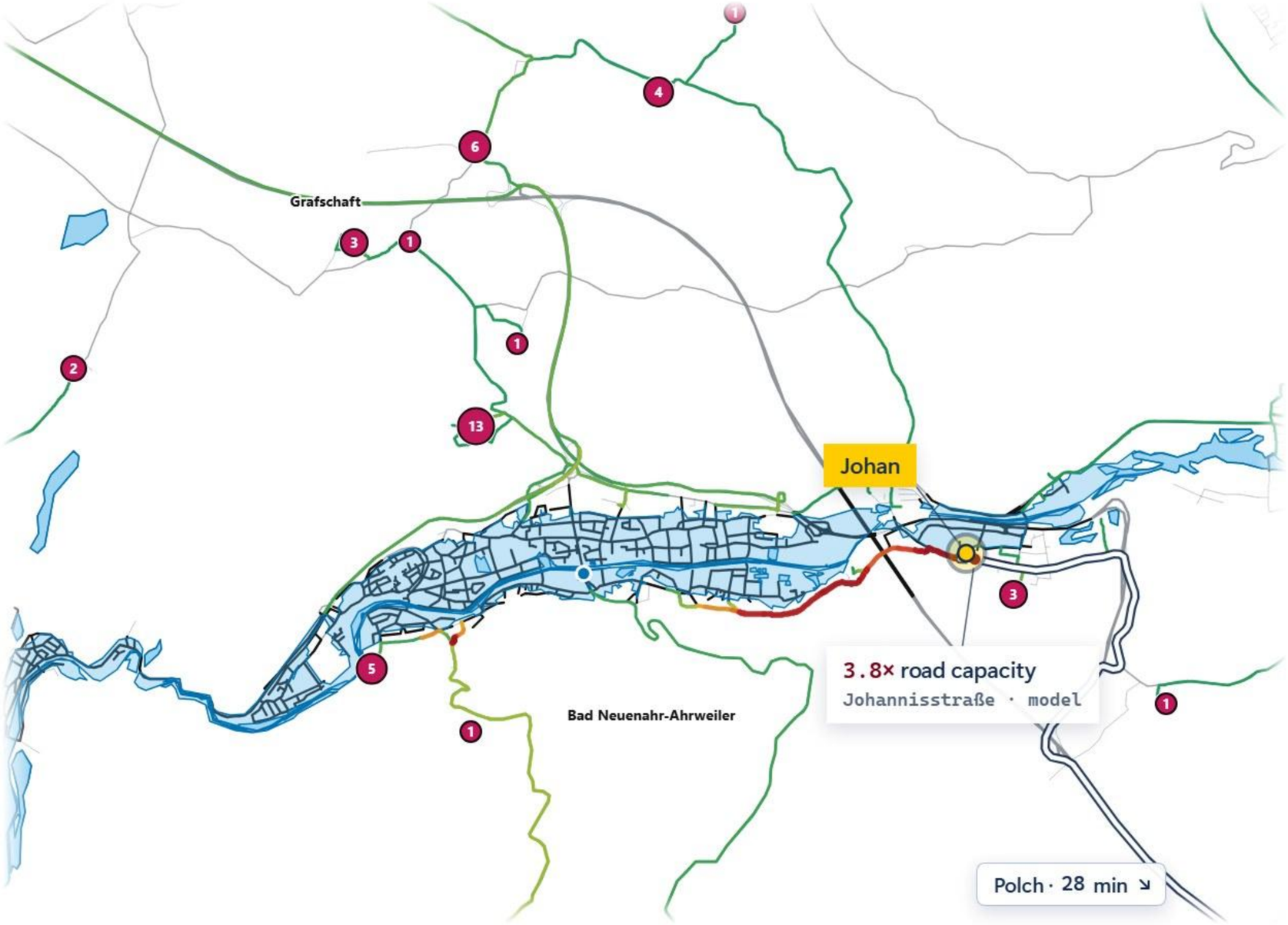
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Core public buildings only

Johan's trip Model, free-flow	28.2 min
Closed roads on driving route	0
Town residents sent to other districts	8,645



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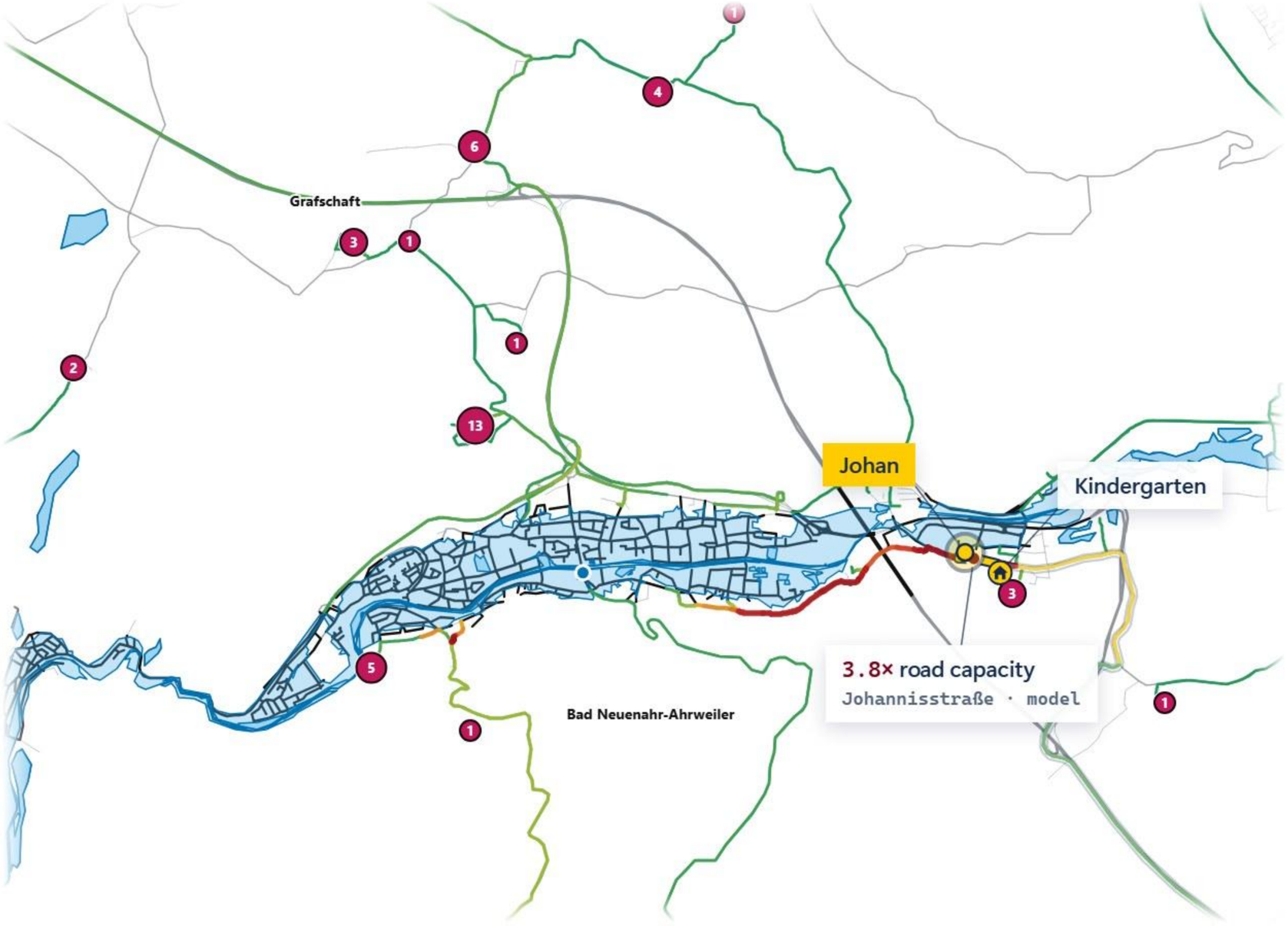
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Streets close. Exits overload.

- Flood, July 2021
- Closed road
- Road stress, model

+ Extended buildings

Johan's trip Model, free-flow	2.4 min
Closed roads on driving route	0
Town residents sent to other districts	0



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Decide before the night.

FLOOD LEVEL

2021 (MAPPED) · 10-yr · 100-yr · 500-yr (MODEL)

SHELTER BUILDINGS

Core public · + Extended

78% reach a shelter within 30 min



within 30 min **19,920** · over 30 min **5,639** · no place **0**

ROADS OVER CAPACITY **37** · MEAN TRIP **22.7** MIN · RESIDENTS **25,559**

Kreis Euskirchen + Landkreis Ahrweiler · residents inside the flood: GHS-POP 2020 estimate · buildings: OpenStreetMap candidates, not designated shelters · Core public = set A, 5 m² per person; + Extended = set B, 3.5 m² per person (assumptions) · Smart Shelter System bundle 2026-10-30

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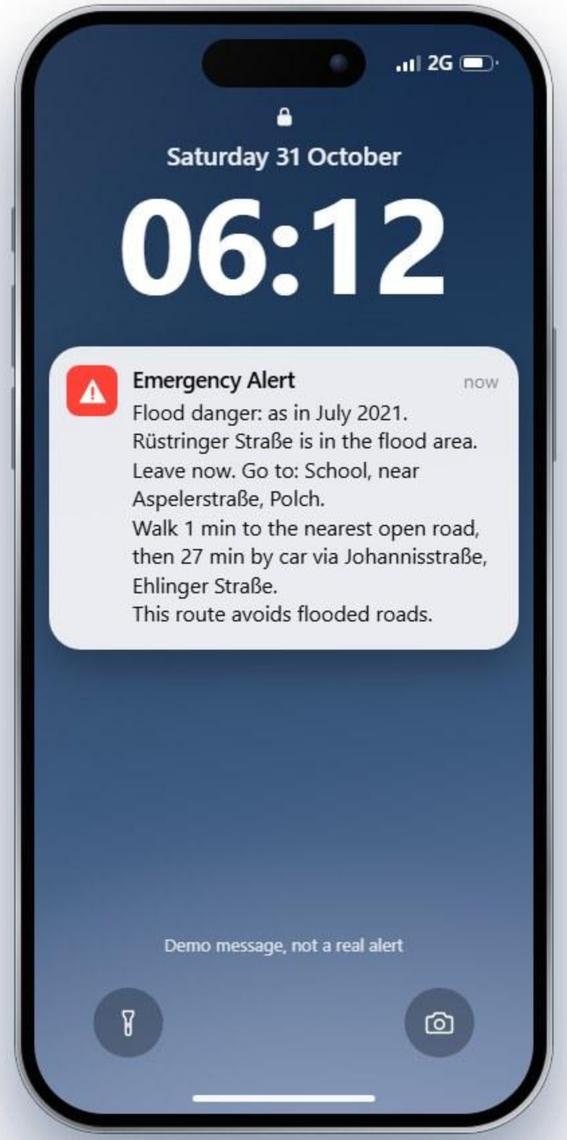
28.2 min

walk 1.3 · car 27.0

Polch

within 30 min **19,920** · over 30 min **5,639** · no place **0**

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Next time, Johan knows where to go.



~230

lives lost, roughly

€32-53 bn

estimated damage

~10×

WMO estimate: early warning returns about 10× its cost



Built on open data:

Copernicus EMS · EFAS · GHSL · OpenStreetMap

July 2021 floods in BE, DE and NL. Deaths and damage: team policy brief refs [2], [9], [20]; return on early warning: WMO 2022, ref [24]. Figures from the team policy brief, not independently verified here.

TO CRISIS AUTHORITIES IN BE · DE · NL

1 **Designate and publish flood shelters.**

2 **Make every alert answer three questions.**

3 **Drill the route at school.**

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TEAM 1

European AI Challenge 2026